

INSTALLATION AND MAINTENANCE INSTRUCTION PILOT OPERATED DIAPHRAGM TYPE SOLENOID VALVE (NORMALLY CLOSE)-PCN SERIES

❖ DESCRIPTION

- Pilot operated (USD type) valves are 2-way normally close solenoid valves designed for low and high pressure service. Valves are made from stainless steel 304 / 316.
- Pilot operated (USD type) valves are also provided with a whether proof and flameproof coil.

IMPORTANT :- Refere other detail from valve catalog for normally open valve.

❖ TEMPERATURE LIMITATIONS

- Circumference Temperature :- - 40 °C to 100 °C
- Media Temperature :- -30 °C to 90 °C

❖ POSITIONING

- Valve is designed to perform properly when mounted in any position. However, for optimum life and performance, the solenoid should be mounted vertical to reduce the possibility of foreign matter accumulating in the solenoid base sub-assembly area.

❖ STORAGE OF VALVES

- On receipt, check the valve to ensure that it is in fully assembled condition.
- Valve should be stored above the ground level or rack.
- Do not apply tar, paint, grease or any other material inside the valve or on plunger as this could impair performance of the valve.

❖ OPERATION

- Pilot operated (USD type) valves have an open and close operation, closing in de-energized position and opening in energized position.
- Diaphragm provides sealing with body orifice and it is held on position by spring compression which is mounted on cover.
- Valve open and closed position working by piston movement in plunger.
- Seat testing/ Body test carried out with hydro and pneumatic test.

❖ INSTALLATION

- If required any compound or chemical apply on male thread only (On pipe). Avoid pipe strain by properly supporting and aligning piping. When tightening the pipe, do not use valve or solenoid as a lever. Locate wrenches applied to valve body or piping as close as possible to connection point.
- To protect the solenoid valve, install a strainer or filter suitable for the service involved in the inlet side as close to the valve as possible. Clean periodically depending on service conditions.
- Check identity sticker for correct valve number, pressure, voltage. Never apply incompatible fluids or exceed pressure rating of the valve.

- Must check coil voltage on coil sticker before connect with power supply.**



Must use flow direction "IN" logo for inlet pipe fitting.



Use Teflon tape for proper joint with valve. (Refer fig. for proper use of Teflon tape)

❖ MAINTENANCE

WARNING: if found any guilty in valve, turn off electrical power, depressurize valve, and vent fluid to a safe area before servicing the valve.

❖ PREVENTIVE MAINTENANCE

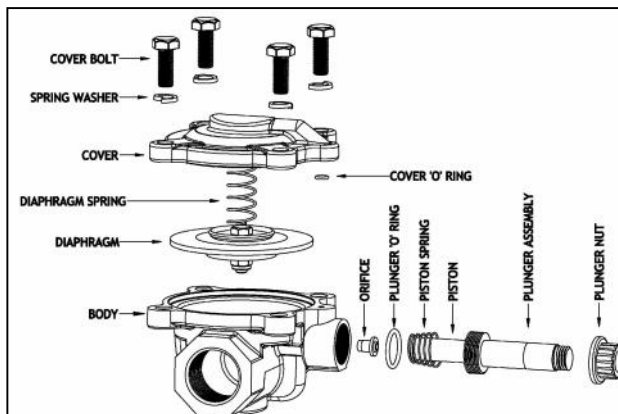
- Keep the medium flowing through the valve as free from dirt and foreign material as possible.
- Valve should be operated/services at least once a month as per your application or usage.

❖ CLEANING

- Valve should be serviced as per application and usage. If not done excessive noise or leakage will be observed. Clean strainer or filter when cleaning the valve.

❖ VALVE DISASSEMBLY

- Remove coil from plunger.
- Remove plunger from body.
- Disassembly of cover from body.
- Remove body diaphragm from valve.
- For normal maintenance, it is not necessary to remove or disassemble the manual operator unless external leakage is evident. If required, remove stem pin and stem with gasket from valve body.(if valve with manual operating)
- All parts are now accessible for cleaning or replacement. If parts are worn or damaged, inform to UFLOWAUTOMATION.



❖ VALVE RE-ASSEMBLY

- Parts must be clean properly.
- Install body diaphragm into valve body.
- Install cover assembly with diaphragm spring into valve body.
- Assemble cover with body by fasteners.
- If removed, re-assemble piston, piston spring and plunger
- Install plunger with piston assembly.
- Recheck line pressure and electrical power supply to valve.
- After reassemble is completed, operate the valve a few times to be sure of proper operation.
- For More detail see figure.



Do not use rusty & never elongate or shorten spring.



Must read name plate specification before installation.



At re-assembly time must clean counter hole of plunger with respectively dia. wire.



Do not use rusted body/damage orifice.

Troubleshooting Guide for Solenoid Valves	
PROBLEM	PROCEDURE
Valve fails to operate	1. Check electrical supply with voltmeter. Voltage must agree with nameplate rating
	2. Check coil with ohmmeter for shorted or opened coil. If open or short replace it.
	3. Make sure that pressure & electric parameter complies with nameplate rating.
Valve is sluggish or inoperative—electrical supply and pressure check out.	1. Disassemble valve; clean out foreign material.
	2. The piston must be free in plunger without binding. Piston spring must be assembled properly without elongate or cut-out length. If damage found then replace it.
	3. Check the diaphragm or diaphragm pad for tears, bulged and/or clogged or obstructed bleed hole or orifice. Damage diaphragm must be replace it.
	4. Check all springs. If broken, replace.
External leakage at flange or joint between body and cover	1. Check that the cover screws are torqued to specifications. If leakage persists, replacement of diaphragm assembly may be required and/or bodies or covers with damaged sealing surfaces may have to be replaced. Flanges are tight as per specification.